

Zhalok Rahman

Software Engineer 2 at Datatree Technology

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Video Introduction: <https://www.loom.com/share/c5ab46a4737244edafd9f052555767a1>

Experience

Kinetik Healthcare

Software Engineer 2

2025-04-01 – Present

- Enhanced realtime notification pipeline with push based mechanism with **Redis Pub/Sub**, **CDC**, **Redis Streams**
- Migrated legacy code from **Express JS** to **Nest JS** for cleaner domain driven design
- Enhanced existing **RBAC** system
- Investigated production incidents based on logs and traces from **CloudWatch** and **Grafana**
- Built **Florence** an **AI voice assistant** which handles around 80 percent of our ride initiation requests
- Developed internal **VoIP** testing tools for simulating calls for the voice agent
- Built a source control pipeline to manage **11Labs agent as code**
- Managed **CI/CD pipeline** for deploying agent
- Integrated Intercom based human agent call center with AI call center for transferring calls as fallback
- Built **Grafana dashboard** for monitoring the AI agent performance
- Contributed in **Langgraph** based orchestrator service for a large scale **AI Agent**

PropertyFinder

Software Engineer

2024-10-01 – 2025-10-31

- Managed reliability of **Golang** based B2B platform
- Monitored **AWS** resource and infrastructure
- Collaborated with data team on data discrepancy issues during migration
- **Investigated and fixed critical production incidents** based on metrics, logs and traces in **Grafana**
- Writen **Golang** scripts for data fix and migration
- Built new features and enhanced existing features with **Golang**, **Lambda**, **ECS**
- Built async pipelines with **SQS**, **Kinesis Streams**, **Step functions**
- Performed **load testing** and monitored the metrics in **Datadog** dashboard for new feature implementations
- Set alerts in **Grafana** with **slack integration**

Learnyx AI

Fullstack Engineer

2023-10-01 – 2025-02-01

- Built API service with **Nest JS**
- Built service for handling AI heavy workload with **FastAPI**, **Langchain**
- Offloaded heavy generation workload to asynchronously with ***RabbitMQ***
- Integrated **Langfuse** for LLM observability
- Built front-end with **React JS**, **Next JS**, **Vite**, **TanStack Query**
- Participated in business decisions from technical point of view with business stakeholders
- Mentored a small team of 2-3 people on different architectures and best practices
- Managed containerized self hosted infrastructure in **AWS EC2** with **Cooffify**
- Built a locally hosted infrastructure with **Docker** exposed with **Ngrok** for staging integration testing

Publications

Optical Mark Recognition with Object Detection and Clustering

2024 6th International Conference on Electrical Engineering and Information and Communication Technology (ICEEICT)

May 2024

<https://doi.org/10.1109/ICEEICT62016.2024.10534561>

- Writen scripts for generating synthetic training data
- Built training scripts with ultralytics for training lightweight YOLO models
- Implemented clustering based unsupervised algorithm for column detection and serialization of answers
- Exported fintuned model in Onnx format for inferencing via API
- demo: <https://www.youtube.com/watch?v=BiEyE5qQqw>

Projects

Remote Browser Control

<https://github.com/zhalok/remote-browser-control>

- Built FastAPI app for receiving commands for playwright to apply on browser session via websocket and emit current snapshot of the browser
- Created a front end app with react to render the remote browser session and also send commands via websocket
- Demo: <https://www.loom.com/share/04b79ed90e344c94a2c50a5b790cda73>

Tech: Playwright, Python, FastAPI, WebSocket

Go Http From Scratch System Calls

<https://github.com/zhalok/go-http-from-scratch-system-calls>

- Built http protocol from sockets and systemcalls
- Handled graceful connection closing

Tech: Golang, Syscalls, OSSignals, Sockets

AI Product Search

<https://github.com/zhalok/ai-product-search>

- Built async vector indexing pipeline from csv file with RabbitMQ and Qdrant
- Built inferencing and searching pipeline with FastAPI
- Used finetuned self hosted SBERT model for generating embedding

Tech: FastAPI, VectorDB (Qdrant), RabbitMQ, SBERT

Certifications

Fundamentals of Operating Systems

Udemy

<https://www.udemy.com/certificate/UC-ff0d44a3-8917-40cd-aef9-a01e258aee49/>

Learned about Process structure, Memory management, CPU cycles, I/O management

NodeJS Internals and Architecture

Udemy

<https://www.udemy.com/certificate/UC-2068cbe0-b17c-47ca-b298-d8acb94b30e2/>

Learned about internal architecture of node js, event loop, node js worker threads, enhancing node js app performance